Mason Miller

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Education

University of Michigan — **Ann Arbor, MI**
Bachelor of Science in Computer Science — *August 2022 – May 2026*

- **GPA:** 3.881/4.000 | **Awards:** William J. Branstrom Freshman Prize, James B. Angell Scholar, University Honors

Publications

Y. Jiang, C. Liu, J. Rose, **M. Miller**, R. Thavamani, J. Wu, Y. Huang, J. Xu, P. Huang. "DistDebug-Bench: Can LLM Agents Diagnose Bugs in Distributed Systems?" *NeurIPS 2026, Evaluations & Datasets Track* (poster).

Y. Jiang, Y. Huang, C. Liu, J. Rose, J. Wu, **M. Miller**, R. Thavamani, P. Huang. "Cirdan: Structured Deep Investigation for Long-Horizon Code Diagnosis." *Under review, ICLR 2027*.

Technical Skills

Languages: C++, Python, C, Go, Bash, SQL

Systems: Linux, XNU/IOKit, POSIX, pthreads, virtual memory, mmap, custom allocators (dlmalloc), io_uring, libaio, NVMe, RocksDB

Distributed: Paxos, Raft, sharding, gRPC, TCP/IP, Docker, Kubernetes, AWS, GCP | **Tools:** GDB, Valgrind, ASan, QEMU, Claude Code

Experience

Apple Inc — **Cupertino, CA**
Software Engineer – Siri Runtime Systems and Interaction — *Starting October 2026*

Citadel LLC — **New York, NY**
Software Engineer Intern – Electronic Trading Engineering Core — *June 2026 – August 2026*

- **Internal Data Platform:** Built a distributed service exposing shared relational data to trading applications on managed container infrastructure, with tenant isolation, per-dataset access control, append-only revision history, and multi-party edit approval
- **Application State Persistence (prototype):** Designed a library restoring in-memory state from a persisted snapshot instead of replaying a week-long event log on crash, using a fixed pre-mapped arena and custom persistent allocator so pointers survive restarts
- Snapshotted without stalling the hot path by capturing a consistent read-only view of memory and writing it asynchronously, validating restores against integrity and version metadata

Ordered Systems Lab, University of Michigan — **Ann Arbor, MI**
Research Assistant (Advisor: Prof. Ryan (Peng) Huang) — *May 2024 – May 2026*

- Co-authored DistDebug-Bench (NeurIPS 2026), a benchmark of 120 real-world bugs from 11 production distributed systems (e.g., ZooKeeper, Ray) for evaluating LLM agents on code-level root-cause diagnosis; curated and annotated failures
- Built Docker-based test harnesses and context-injection pipelines for Cirdan (under review), an LLM agent harness for diagnosing distributed-system failures; separately benchmarked async I/O (libaio, io_uring) under RocksDB across queue depths and workloads

Apple Inc — **Cupertino, CA**
Software Engineer Intern – CoreOS Storage Device Drivers — *May 2025 – August 2025*

- Enabled user-space testing of kernel storage drivers without physical hardware probing by building a C++ tool that fakes the XNU IOKit framework, with a fake storage device, automatic driver matching, and error injection into mock block devices
- Raised driver code coverage by over 80% with the new harness and used it to expose a 15-year-old bug in the XNU kernel

Qumulo Inc — **Seattle, WA**
Member of Technical Staff Intern – Cloud Read Cache — *January 2025 – April 2025*

- Cut minimum cluster cost by 33% by designing and deploying single-node clusters with disk-level fault tolerance, and built automated AWS/Azure performance testing to benchmark single- vs. multi-node configurations
- Led a distributed key/value read cache upgrade with safe key remapping, quorum barriers, and RPC-based shard coordination; instrumented cache hits, misses, and gap-fills to optimize the read path

EECS Staff, University of Michigan — **Ann Arbor, MI**
Instructional Assistant – Distributed Systems (W26), Operating Systems (F25) — *August 2025 – May 2026*

- Taught Paxos, replication, and sharding (EECS 491); threads, virtual memory, and file systems (EECS 482); 5–10 office hours/week

Projects

Sharded Key/Value Service with Paxos Groups (Go)

- Built a fault-tolerant sharded key/value store with shard migration across Paxos-replicated groups, a shard master for configuration and load balancing, and atomic two-phase handoff that guarantees linearizability across reconfiguration

Multi-threaded Network File Server (C/C++)

- Built a concurrent network file server over TCP serving many simultaneous users across nested directories, using fine-grained reader-writer locking with hand-over-hand traversal and ordered disk writes preserving metadata invariants